

TRUNG TUAN MAI

trungtuan.mai@ucalgary.ca
linkedin.com/in/tuanmai3011

Cell: (825) 488-2472

EDUCATION

Bachelor of Science, Software Engineering Schulich School of Engineering, University of Calgary	Expected April 2027
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CORE SKILLS

Programming: Python, Java, JavaScript
AI / ML: scikit-learn, feature engineering, regression modeling, model evaluation
Data: pandas, NumPy, data preprocessing, pipeline design
Systems: Flask, REST APIs
Tools: Docker, Git, Linux
Concepts: machine learning pipelines, data pipelines, model evaluation, system integration

PROJECTS

Predicting Car Sales Revenue – Machine Learning System (Machine Learning Course)	December 2025
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- Engineered an end-to-end machine learning pipeline in Python using pandas, NumPy, and scikit-learn to process a dataset exceeding **500K records**.
- Designed structured preprocessing workflows using ColumnTransformer, StandardScaler, and OneHotEncoder to ensure consistent feature transformation and prevent data leakage.
- Engineered 15+ predictive features through statistical correlation analysis and temporal feature extraction to improve model performance.
- Evaluated multiple regression models across 47+ controlled experiments, achieving **R² = 0.91** and selecting optimal models based on RMSE and MAE metrics.
- Packaged the final model pipeline using joblib to support reproducible inference on unseen data

AI / NLP Risk Intelligence System (Self-Directed Project – In Progress)	In Progress
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- Designed an NLP-based system architecture for extracting structured risk insights from unstructured documents using LLM-assisted pipelines.
- Planned retrieval and reasoning workflows integrating document parsing, semantic retrieval, and structured database querying.
- Defined API-driven architecture to support automated risk classification and multi-step information synthesis.

Data Ingestion & Backend System (Web-Based Systems Course)	February 2026
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- Engineered a Flask backend with REST APIs to ingest and normalize multi-source datasets into MongoDB for downstream analytics.
- Implemented schema validation and unique-key deduplication to ensure consistent and reliable training data inputs.
- Designed an ETL-style pipeline combining scraping, parsing, transformation, and persistence to automate data collection workflows.

EXTRA-CURRICULARS

Financial Transaction Processing Backend JP Morgan Chase Virtual Experience in Forage	January 2026
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- Developed Spring Boot backend services exposing REST endpoints for data retrieval and persistence, supporting integration with external systems.
- Integrated Kafka-based event listeners to simulate asynchronous data processing pipelines in an event-driven architecture.

WORK EXPERIENCE

Server - Pho Pham Restaurant	Present
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- Coordinated high-volume operations handling 100+ transactions per shift while maintaining workflow efficiency.
- Resolved real-time operational issues through structured problem-solving in a fast-paced environment.